

HAT-P-20b

R-0998

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_251223 · created 2026-07-30T01:00:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461032.8291 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.053
Transit depth (Rp/Rs)^2	1.49 +/- 1.3 [%]
Orbital Inclination (inc)	89.56 +/- 2.4
Airmass coefficient 1 (a1)	1.474 +/- 0.023
Airmass coefficient 2 (a2)	-0.332 +/- 0.052
Scatter in the residuals of the lightcurve fit is	1.58 %
Best Comparison Star	#1 - [195, 382]
Optimal Aperture	3.94
Optimal Annulus	10.86
Transit Duration (day)	0.054 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.122 $\pm$ 0.053 vs 0.1284 (deviation 0.11 σ)
Duration fit vs published	0.054 d vs 0.0775 d (deviation 1.87 σ)
Mid-time O-C	-8.9 min
Depth SNR	2.0
Residual scatter	1.58 %

Light curve

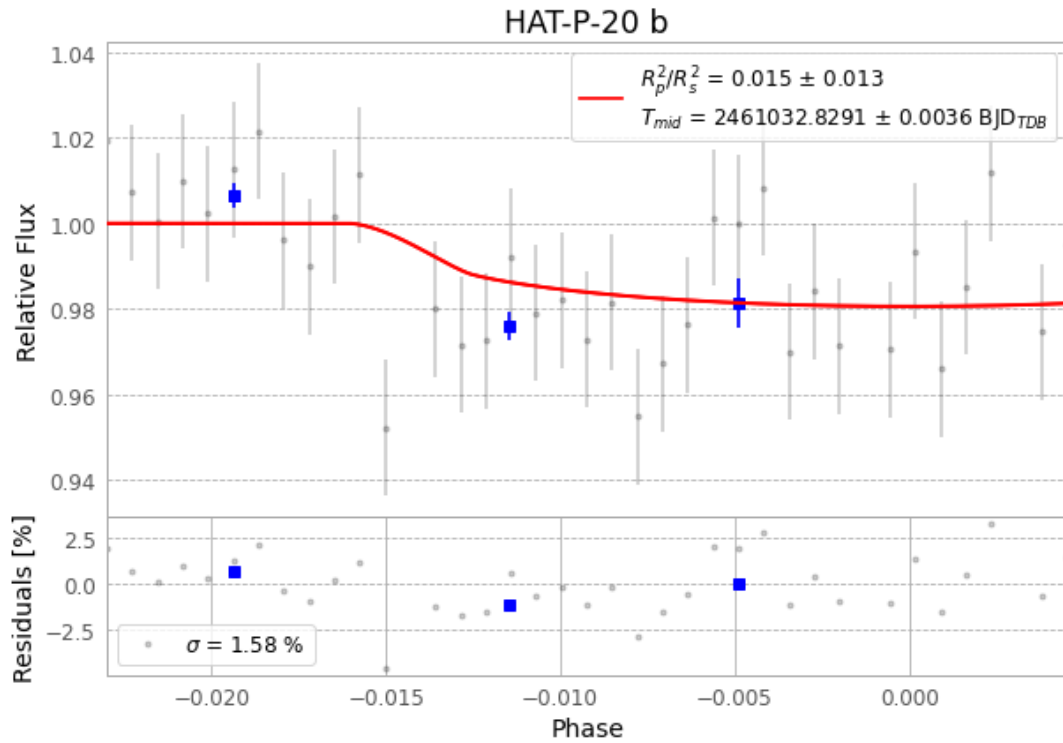


Figure 1 — FinalLightCurve_HAT-P-20 b_2025-12-22.png (SHA-256 49312aa8e326ed0c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 329d6f9c5d05569d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

39 FITS frames (26 MB), HATP-20251223060916.FITS → HATP-20251223080316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-251223.log (d70658599697...), FinalLightCurve_HAT-P-20 b_2025-12-22.png (49312aa8e326...), FinalLightCurve_HAT-P-20 b_2025-12-22.csv (f7904d51fd14...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 01:00Z.